

Lab Evaluation:

Course Name: Object Oriented Programming				Course Code: CMPE221L		
Assignment Type: Lab				Dated: 16 February 2026		
Semester: 2 nd				Session: CYS 2025 Spring		
Lab/Project/Assignment #: 1				CLOs to be covered: CLO1		
Lab Title: Getting Started				Teacher Name: Mr. Mujtaba		
CLO 1	Implement abstraction and encapsulation to develop reusable classes for objects of real-world problems					
Levels (Marks)	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6
(10)						
Total						/10

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	9-10	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	7-8	L2	Submitted the lab tasks but have good understanding
Good	5-6	L3	Submitted the lab tasks but have weak understanding.
Basic	3-4	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1-2	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No.1

Lab Goals/Objectives:

Getting Started with Java

Acquiring practice with familiar programming constructs in JAVA

Equipment Required: Intel J IDEA Edition Community, Net Beans with JDK 13 Version, Visual Studio.

Theory/Relevant Material: Write First Java Program

Lab Tasks:

1. Change the *Factorial* code discussed in class so that it shows proper messaging in case of negative numbers.
2. Change the *Series_Generator* such that it can generate both increasing series or decreasing series depending on the starting and ending values.
3. Write a Program that takes year as an input from user and determine if it's a leap year or not.

Sample input

>>>

Enter Any Year: 1940

It is a leap year

4. You are interested in calculating your electricity bill of your home. You know the unit consumption of your home. Billing criteria for electric company is that it charges 5 Rs/ unit for first 100 units, next 100 units are charged at 7 Rs/unit and above that charge is 15 Rs/unit. As input, you get **total_Units** consumed and generate bill.

Sample input

>>>

Enter Units Consumed: 150

Bill: 850

5. Write a program that takes an integer n from command-line arguments and uses a while loop to compute the number of times you need to divide n by 2 until it is strictly less than 1. Print the error message "Illegal input" if n is negative.

Sample input

>>> n: 8 Div: 4

>>> n: 4

Div: 3

6. Write a code that takes a number value from user and displays the inverse of it. Number can be of any length.

Sample input

>>>

Number: 151

Inverse: 151

>>>

Number: 1567

Inverse: 7651